IN DIFFERENTIALLY CLOSED FIELDS THE NON-MINIMALITY DEGREE AGREES WITH THE FIBRE DIMENSION

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JB was partially supported by Simons travel grant G3535. JF was partially supported by NSF CAREER award number 1945251. SM, who is additionally affiliated with the Institut de mathématiques de Jussieu – Paris Rive Gauche, was supported by the NSF under Grant No. DMS-2303034, and by the European Union's Horizon 2020 research and innovation programme under the Marie Sklodowska-Curie grant agreement No 101034255.

Editorial note. The counted proof contains one reference to another labelled item, the remark labelled findbigger, recorded as a navigation stub of kind remark.

Allowed standard prerequisites (editorial).

- The theory DCF_0 of differentially closed fields of characteristic zero, and the two invariants compared: the non-minimality degree and the fibre dimension F_ind. Set up in the paragraph preceding the counted statement (source lines 440-441). Fixes the theory and the two quantities whose equality is asserted.
- The classification of types with no proper fibrations into locally modular and non-locally-modular cases. Cited inside the counted proof (Moosa, Proposition 2.3) and used at lines 446-451. Supplies the dichotomy that splits the proof into its two cases.
- Knop's classification of multiply transitive algebraic group actions. Cited inside the counted proof at line 455. Identifies the binding group action with affine transformations or the action of PGL on projective space, which pins the fibre dimension to 2.

Source reference. Original remark 2.1.5: sources/logic-fl03-simple-homogeneous-structures-and/pinned-original-source.tex, source lines 451–451. Navigation pointer for \ref{findbigger} used inside the counted span (line 451); the argument that consumes it is retained in the span.

Proposition 0.1. For a type p in the theory DCF_0, $\mathrm{nmdeg}(p) = F_{ind}(p)$.

Proof. Let $p \in S(A)$. If $p_n \to p_{n-1} \to \ldots \to p_1$ is an analysis of the type p over A, $F_{ind}(p)$ is bounded by $F_{ind}(p_1)$ and also F_{ind} of the types given by the generic fibers of the maps in the analysis. Thus, it suffices to prove the result in the case that p has no proper fibrations [3, Definition 2.1]. It follows by [3, Proposition 2.3] that p is either

(1) interalgebraic over A with a finite collection of independent realizations of a locally modular minimal type, or
(2) interalgebraic with a type q which is internal to a nonlocally modular type.

In the first case, both $F_{ind}(p) = \mathrm{nmdeg}\,(p) = 1$. In the second case, if the binding group of p relative to the constants is not generically 2-transitive [1, for the defintion of generic transitivity], then $F_{ind}(p) = 1$, and $\mathrm{nmdeg}(p) \leq F_{ind}(p)$ in general (see Remark 2.1.5), so the quantities are equal. In the second case, since $F_{ind}(-)$ is invariant up to interalgebraicity, we can assume p is internal to the nonlocally modular type - in this case, the constants.

Suppose now that $F_{ind}(p) > 1$. Then it must be that p and also $p \otimes p$ are weakly orthogonal to the constants over A (if not, it is easy to see that $F_{ind}(p) = 1$ by taking a Morley sequence whose image in $\mathcal{C}^n$ coming from nonorthogonality lies on an algebraic curve). It follows that both p and $p \otimes p$ are isolated types. We wish now to reduce to the case that the action of the binding group G on p is 2-transitive. This is equivalent to any two realizations of p being independent. As the binding group is generically 2-transitive, with $p \otimes p$ being isolated, there must be an A-definable equivalence relation $\sim$ on p such that $p/\sim$ has a 2-transitive binding group.

But now the binding group is 2-transitive and definably isomorphic to an algebraic group action acting on an algebraic variety over a field of characteristic zero, and so the classification theorem of Knopp [2] tells us that the binding group action on p is isomorphic to either a group of affine transformations of affine space or the action of PGL_{n+1} on $\mathbb{P}^n$. Here it must be that $n > 1$. Both of these group actions preserve collinearity of points. Taking the Morley sequence of points on a line, we

can see that $F_{ind}(p) = 2$. $\mathrm{nmdeg}(p) = 2$ in this case as well since the binding group acts 2-transitively. □

References

[1] James Freitag and Rahim Moosa. Bounding nonminimality and a conjecture of borovik–cherlin. Journal of the European Mathematical Society, 2023.

[2] Friedrich Knop. Mehrfach transitive operationen algebraischer gruppen. Archiv der Mathematik, 41:438–446, 1983.

[3] Rahim Moosa and Anand Pillay. Some model theory of fibrations and algebraic reductions. Selecta Mathematica, 20(4):1067–1082, 2014.